

Nexus Broadcast — Qt 6 Widgets Developer Handoff

Target: Native C++ application using Qt 6 Widgets + QSS (Qt Style Sheets). No QML. No web technologies. **Prototype reference:** React/Tailwind prototype at [nexus-broadcast-ui/](#) (see attached ZIP). **Instruction:** Document every measurement, color, and behavior exactly as implemented. Do not simplify or redesign.

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1. Design Token Reference

All colors are defined here. Use these exact values everywhere. Do not approximate.

1.1 Surface Colors

Token Name	Hex / RGBA	Qt QColor Constructor	Usage
surface-root	#0E0F14	QColor(14, 15, 20)	App root background, main content area
surface-topbar	#0D0E12	QColor(13, 14, 18)	Top bar background (all screens)
surface-sidebar	#111318	QColor(17, 19, 24)	Sidebar background
surface-panel	#111318	QColor(17, 19, 24)	Panel / card backgrounds
surface-elevated	#16181F	QColor(22, 24, 31)	Settings tab content area, elevated panels
surface-input	#0E0F14	QColor(14, 15, 20)	Text inputs, selects, number fields
surface-active	#1A1D2B	QColor(26, 29, 43)	Selected nav item, selected source row
surface-canvas	#0A0B0F	QColor(10, 11, 15)	Program output canvas, scene editor workspace
surface-channel	#0E0F14	QColor(14, 15, 20)	Individual audio channel cards
surface-chroma-card	#0E0F14	QColor(14, 15, 20)	Chroma key sub-panel

1.2 Border Colors

Token Name	RGBA	Qt QColor Constructor	Usage
border-primary	rgba(255,255,255,0.07)	QColor(255,255,255,18)	Panel edges, sidebar border
border-panel	rgba(255,255,255,0.08)	QColor(255,255,255,20)	Card/panel borders
border-section	rgba(255,255,255,0.06)	QColor(255,255,255,15)	Inner section dividers within panels
border-subtle	rgba(255,255,255,0.05)	QColor(255,255,255,13)	FormRow bottom dividers
border-input	rgba(255,255,255,0.12)	QColor(255,255,255,31)	Text inputs, selects
border-input-focus	rgba(59,130,246,0.5)	QColor(59,130,246,128)	Input focus ring
border-divider	rgba(255,255,255,0.10)	QColor(255,255,255,26)	Vertical dividers in top bar
border-ghost	rgba(255,255,255,0.08)	QColor(255,255,255,20)	Inactive scene card, inactive tool button

1.3 Accent Colors (Semantic — Never Swap)

Token Name	Hex	Qt QColor	Semantic Meaning
accent-blue	#3B82F6	QColor(59,130,246)	Primary interaction, active states, CTAs, nav active border
accent-blue-bg	rgba(59,130,246,0.10)	QColor(59,130,246,26)	Blue chip/badge background
accent-blue-border	rgba(59,130,246,0.25)	QColor(59,130,246,64)	Blue chip/badge border
accent-blue-glow	rgba(59,130,246,0.30)	QColor(59,130,246,77)	Active scene card box-shadow
accent-red	#EF4444	QColor(239,68,68)	LIVE badge, End Stream button, critical alerts
accent-red-bg	rgba(239,68,68,0.15)	QColor(239,68,68,38)	LIVE badge background
accent-red-border	rgba(239,68,68,0.35)	QColor(239,68,68,89)	LIVE badge border
accent-green	#22C55E	QColor(34,197,94)	Connected, healthy, SRC ACTIVE, CPU/GPU good
accent-amber	#F59E0B	QColor(245,158,11)	Warning, degraded bitrate, donation highlight
accent-cyan	#06B6D4	QColor(6,182,212)	Timecodes, uptime counter, secondary data
accent-purple	#9146FF	QColor(145,70,255)	Twitch platform color (platform selector only)
accent-youtube	#FF0000	QColor(255,0,0)	YouTube platform color (platform selector only)
accent-facebook	#1877F2	QColor(24,119,242)	Facebook platform color (platform selector only)

1.4 Text Colors

Token Name	RGBA	Qt QColor	Usage
text-primary	#FFFFFF	QColor(255,255,255)	Active nav item, selected source, KPI values
text-secondary	rgba(255,255,255,0.60)	QColor(255,255,255,153)	Inactive nav items, secondary labels
text-tertiary	rgba(255,255,255,0.40)	QColor(255,255,255,102)	Section headers, form labels
text-muted	rgba(255,255,255,0.35)	QColor(255,255,255,89)	Metadata, timestamps, placeholder
text-dim	rgba(255,255,255,0.25)	QColor(255,255,255,64)	Instrumentation strip labels (TC, SCENE, etc.)
text-ghost	rgba(255,255,255,0.15)	QColor(255,255,255,38)	Placeholder icons, empty state text
text-separator	rgba(255,255,255,0.12)	QColor(255,255,255,31)	Pipe `
text-disabled	rgba(255,255,255,0.20)	QColor(255,255,255,51)	Disabled button labels

1.5 Spacing & Radius Tokens

Token	Value	Usage
spacing-xs	4px	Icon-to-label gap, badge padding
spacing-sm	8px	Gap between controls, inner padding
spacing-md	12px	Panel padding, section gap
spacing-lg	16px	Outer content padding
radius-sm	4px	Badges, chips, small buttons
radius-md	6px	Buttons, inputs, number fields
radius-lg	8px	Panels, cards, scene cards
radius-full	9999px	Toggle switches, health bars, VU bar segments

2. Typography System

Three fonts are required. Embed them as application resources (`.ttf` or `.otf` files) and load via `QFontDatabase::addApplicationFont()` .

Font Family	Weights Required	Role	Qt Usage
Space Grotesk	500, 600, 700	Display / Brand — page titles, KPI values, logo wordmark	QFont("Space Grotesk", size, QFont::Bold)
Inter	400, 500, 600	UI / Body — nav labels, form labels, body copy, button text	QFont("Inter", size)
JetBrains Mono	400, 500, 700	Data / Precision — timecodes, bitrate, dB, FPS, resolution	QFont("JetBrains Mono", size)

2.1 Type Scale

Size	Font	Weight	Color Token	Usage
9px	Inter or JetBrains Mono	400–600	text-muted / text-dim	Micro labels, badge text, version, signal indicators, VU dB readouts
10px	Inter (uppercase)	600	text-tertiary	Section headers (SCENES , AUDIO MIXER , STREAM STATUS), instrumentation strip labels
10px	JetBrains Mono	400	text-muted	Top bar metadata (YT, FPS, resolution, codec, SIG OK)
11px	Inter	400–500	text-secondary	Status text, timestamps, channel sub-labels
12px	Inter	400–500	text-secondary	Form labels, nav items, settings values, source names
12px	Space Grotesk	700	text-primary	Page title in top bar (all-caps, letter-spacing 0.1em)
13px	Inter	500–600	text-primary	End Stream button text, bitrate value label
13px	JetBrains Mono	700	accent-blue	Bitrate kbps value in Stream Status panel
15px	JetBrains Mono	700	text-primary	Viewers count, uptime in stat cards
20px	Space Grotesk	700	text-primary	KPI card values (Analytics screen)
28px+	Space Grotesk	700	text-primary	Hero numbers (peak viewers, revenue)

2.2 Section Header Pattern

All panel section headers (e.g., “SCENES”, “AUDIO MIXER”, “STREAM STATUS”, “SOURCES”) use this exact style:

Font: Inter
Weight: 600
Size: 10px
Color: rgba(255,255,255,0.40) [text-tertiary]
Letter-spacing: 0.1em (approximately 1px tracking per 10px)
Transform: UPPERCASE

In Qt: QLabel with QFont("Inter", 8, QFont::DemiBold) , setStyleSheet("color: rgba(255,255,255,102); letter-spacing: 1px;") .

3. Global Application Shell & Layout

3.1 Window Structure

The application window is a fixed horizontal split: a persistent left sidebar and a right main content area. There is no top-level menu bar or title bar chrome — use Qt::FramelessWindowHint with a custom title bar or QMainWindow with menuBar()→hide() .

```
QMainWindow (min: 1024×706, default: 1440×900)
└─ Central Widget: QWidget
    └─ QHBoxLayout (spacing: 0, margins: 0)
        └─ SidebarWidget (fixed width: 220px)
            └─ MainContentStack (QStackedWidget, flex-1)
                └─ DashboardPage
                └─ SceneEditorPage
                └─ ChatPage
                └─ AnalyticsPage
                └─ SettingsPage
```

3.2 Dimensional Constants

Constant	Value	Description
SIDEBAR_WIDTH	220px	Fixed, never collapses on desktop
TOPBAR_HEIGHT	46px	Minimum height for all page top bars
SIDEBAR_LOGO_HEIGHT	56px	Logo block minimum height
DASHBOARD_RIGHT_PANEL_WIDTH	240px	Stream Status right panel
SCENE_EDITOR_RIGHT_PANEL_WIDTH	320px	Sources + Properties right panel
SETTINGS_TAB_NAV_WIDTH	180px	Settings left tab navigation
CHAT_LEFT_PANEL_WIDTH	220px	Chat left utility rail
CHAT_RIGHT_PANEL_WIDTH	260px	Chat right activity rail
SCENE_CARD_WIDTH	110px	Scene switcher card width
SCENE_CARD_HEIGHT	70px	Scene switcher card height
TOOL_BUTTON_SIZE	32px	Scene editor vertical tool buttons
HEALTH_BAR_HEIGHT	4px	Stream health progress bars
VU_BAR_WIDTH	3px	Individual VU meter bar width
VU_BAR_COUNT	20	Number of bars per VU meter
VU_METER_HEIGHT	16px	Total VU meter container height
MASTER_SLIDER_WIDTH	80px	Master volume slider width
BITRATE_SPARKLINE_HEIGHT	40px	Bitrate SVG sparkline height
TOGGLE_WIDTH	36px	Toggle switch width (Settings) / 32px (Scene Editor)
TOGGLE_HEIGHT	20px	Toggle switch height (Settings) / 18px (Scene Editor)
TOGGLE_THUMB_SIZE	16px	Toggle thumb diameter (Settings) / 14px (Scene Editor)

4. Sidebar Component

4.1 Structure

```
SidebarWidget (QWidget, fixed width: 220px, background: #111318)
├── LogoBlock (QWidget, min-height: 56px, border-bottom: 1px rgba(255,255,255,18))
│   ├── LogoIcon (QLabel, 28×28px, background: #3B82F6, border-radius: 4px)
│   │   └── Radio SVG icon (15×15px, white)
│   ├── BrandStack (QVBoxLayout, spacing: 0)
│   │   ├── "NEXUS" (QLabel, Space Grotesk Bold, 13px, #fff, letter-spacing: 0.04em)
│   │   └── "BROADCAST" (QLabel, Inter Regular, 9px, rgba(255,255,255,102), letter-spacing: 0.12em)
│   └── ProBadge (QLabel, "PRO", 9px Inter Bold, #fff, background: #3B82F6, border-radius: 4px, padding: 0 4px)
├── NavList (QVBoxLayout, top-padding: 8px, bottom-padding: 8px)
│   └── NavItem × 5 (see 4.2)
└── StatusFooter (QWidget, border-top: 1px rgba(255,255,255,18), padding: 12px 16px)
    ├── ConnectionRow (QHBoxLayout)
    │   ├── WiFi icon (13×13px, #22C55E)
    │   └── "Connected · Excellent" (Inter 500, 11px, #22C55E)
    ├── LiveRow (QHBoxLayout, margin-top: 4px)
    │   ├── LiveDot (QLabel, 8×8px, background: #EF4444, border-radius: 4px, animated pulse)
    │   └── "LIVE · 01:23:47" (JetBrains Mono, 11px, rgba(255,255,255,128))
    └── SignalRow (QHBoxLayout, margin-top: 8px, border-top: 1px rgba(255,255,255,15))
        ├── SignalBars (custom QWidget – see 4.3)
        └── "NEXUS v2.4.1" (JetBrains Mono, 9px, rgba(255,255,255,77))
```

4.2 Nav Item States

Each nav item is a `QPushButton` or `QWidget` with mouse event handling. Exact pixel values:

State	Background	Left Border	Text Color	Icon strokeWidth	Padding-left
Default	transparent	3px transparent	rgba(255,255,255,128)	1.5	16px
Hover	rgba(255,255,255,13)	3px transparent	rgba(255,255,255,204)	1.5	16px
Active/Selected	#1A1D2B	3px solid #3B82F6	#FFFFFF	2.0	13px (compensates for border)
Pressed	rgba(255,255,255,20)	3px transparent	rgba(255,255,255,204)	1.5	16px

Nav item internal layout:

- Height: 38px minimum
- Icon: 16×16px, left-aligned
- Gap between icon and label: 12px
- Label: Inter, 13px, weight 400 (default) / 500 (active)
- Transition: background color 120ms ease-out

4.3 Signal Bars Widget

Five vertical bars, rendered as a custom `QWidget::paintEvent`:

Bar index (0–4)	Width	Height	Color (active)	Color (inactive)
0	3px	10px	#3B82F6	rgba(255,255,255,26)
1	3px	12px	#3B82F6	rgba(255,255,255,26)
2	3px	14px	#3B82F6	rgba(255,255,255,26)
3	3px	16px	#3B82F6	rgba(255,255,255,26)
4	3px	18px	rgba(255,255,255,26)	rgba(255,255,255,26)

Bars 0–3 are active (blue), bar 4 is inactive (dim). Gap between bars: 1px. Border-radius: 1px on each bar. Bars are bottom-aligned.

5. Top Bar (Page Header)

Every screen has an identical-height top bar. The content changes per screen but the container is always:

```
TopBarWidget (QWidget)
  height:          46px minimum
  background:      #0D0E12
  border-bottom:   1px solid rgba(255,255,255,18)
  padding:         0 16px
  layout:          QHBoxLayout, spacing: 8px, vertical-center
```

5.1 Top Bar — Dashboard

Left-to-right element sequence with exact spacing:

```
[PAGE TITLE] [DIVIDER] [LIVE BADGE] [PLATFORM] [ ] [TIMECODE] [ ] [RESOLUTION] [ ] [FPS] [ ] [CODEC] ↔
```

Element	Widget	Font	Size	Color	Notes
"DASHBOARD"	QLabel	Space Grotesk Bold	12px	#fff	letter-spacing: 0.1em
Vertical divider	QFrame (VLine)	—	1×16px	rgba(255,255,255,26)	
LIVE badge	Custom QWidget	Inter Bold	10px	#EF4444	bg: rgba(239,68,68,38), border: 1px rgba(239,68,68,89), radius: 4px, padding: 2px 8px, includes pulsing dot
"YT"	QLabel	JetBrains Mono	10px	rgba(255,255,255,77)	Platform abbreviation
Pipe separators	QLabel	JetBrains Mono	10px	rgba(255,255,255,31)	"
Timecode	QLabel	JetBrains Mono	10px	#06B6D4	DYNAMIC — updates every second
"1920×1080"	QLabel	JetBrains Mono	10px	rgba(255,255,255,89)	DYNAMIC — from encoder config
"60fps"	QLabel	JetBrains Mono	10px	rgba(255,255,255,89)	DYNAMIC — from encoder config
"H.264"	QLabel	JetBrains Mono	10px	#22C55E	DYNAMIC — from encoder config
[spacer]	QSpacerItem	—	—	—	pushes right side
SIG OK dot	QLabel	—	6×6px	#22C55E	circle, border-radius: 3px
"SIG OK"	QLabel	JetBrains Mono	9px	rgba(255,255,255,89)	DYNAMIC — from network status
Divider	QFrame	—	1×16px	rgba(255,255,255,20)	
"Default Profile"	QLabel	Inter	10px	rgba(255,255,255,77)	DYNAMIC — from profile name
Settings2 icon	QPushButton (icon-only)	—	13×13px	rgba(255,255,255,77)	Opens profile settings

5.2 Top Bar — Scene Editor

[SCENE EDITOR] [DIVIDER] [PROJECT CHIP] [→ spacer →] [● LIVE 00:00:00] [⏏] [REC 00:00:00] [⏏] [CPU x.x%]

Element	Color	Notes
"SCENE EDITOR"	#fff	Space Grotesk Bold 12px, letter-spacing 0.1em
Project chip	bg: rgba(59,130,246,26), border: 1px rgba(59,130,246,64)	JetBrains Mono 10px, #3B82F6, radius: 4px, padding: 2px 8px
LIVE timecode	#EF4444 dot + rgba(255,255,255,77) text	DYNAMIC
REC timecode	rgba(255,255,255,77)	DYNAMIC
CPU %	label: rgba(255,255,255,77), value: #22C55E	DYNAMIC
FPS	label: rgba(255,255,255,77), value: #3B82F6	DYNAMIC
KB/s	label: rgba(255,255,255,77), value: #06B6D4	DYNAMIC

6. Dashboard Screen — Full Specification

6.1 Layout

```
DashboardPage (QWidget, background: #0E0F14)
├── TopBar (46px, see §5.1)
├── ContentArea (QHBoxLayout, spacing: 0)
│   ├── CenterColumn (QVBoxLayout, padding: 12px, spacing: 12px, flex-1)
│   │   ├── ProgramOutputPanel
│   │   ├── ScenesPanel
│   │   └── AudioMixerPanel
│   └── StreamStatusPanel (fixed width: 240px, border-left: 1px rgba(255,255,255,18))
```

6.2 Program Output Panel

```
ProgramOutputPanel (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px)
├── PanelHeader (QHBoxLayout, height: 32px, padding: 0 12px, border-bottom: 1px rgba(255,255,255,15))
│   ├── "PROGRAM OUTPUT" label (section header style)
│   └── LIVE badge (right-aligned)
├── CanvasArea (custom QWidget, aspect-ratio: 16:9, background: #0A0B0F)
│   ├── GridOverlay (SVG-style grid, opacity: 6%, see §6.2.1)
│   ├── PlaceholderContent (centered Monitor icon + resolution string)
│   └── CornerMarkers × 4 (see §6.2.2)
└── InstrumentationStrip (QWidget, height: 28px, background: #0A0B0F, border-top: 1px rgba(255,255,255,13))
    ├── LeftGroup: TC [timecode] | SCENE [name] | SRC ACTIVE
    └── RightGroup: VIEWERS [count] | BITRATE [kbps]
```

Canvas aspect ratio enforcement: Use `QWidget::heightForWidth()` returning `width * 9 / 16`, or constrain with a `QSizePolicy` with `setHeightForWidth(true)`.

6.2.1 Canvas Grid Overlay

Render in `paintEvent` using `QPainter`:

- Grid cell size: 40×40px
- Line color: `QColor(255,255,255,15)` (opacity ~6%)
- Line width: 0.5px (use `QPen` with `setWidthF(0.5)`)
- Pattern: horizontal + vertical lines at every 40px interval
- Rendering: `QPainter::Antialiasing` off for crisp pixel lines

6.2.2 Corner Bracket Markers

Four L-shaped brackets, one at each corner of the canvas. Render in `paintEvent` :

- Size: 16×16px each
- Color: `QColor(59,130,246,153)` (`rgba(59,130,246,0.6)`)
- Line width: 1.5px
- Position: 8px inset from each corner edge
- Shape: top-left = horizontal right + vertical down; top-right = horizontal left + vertical down; bottom-left = horizontal right + vertical up; bottom-right = horizontal left + vertical up

6.2.3 Instrumentation Strip

Height: 28px. Background: `#0A0B0F` . Border-top: 1px `rgba(255,255,255,13)` . Padding: 0 12px.

Left group (JetBrains Mono, 9px):

- “TC” label: `rgba(255,255,255,64)` [text-dim]
- Timecode value: `#06B6D4` [accent-cyan] — **DYNAMIC**, format `HH:MM:SS:FF`
- “|” separator: `rgba(255,255,255,26)` [text-separator]
- “SCENE” label: `rgba(255,255,255,64)`
- Scene name: `#FFFFFF` — **DYNAMIC**
- “|” separator
- “SRC ACTIVE”: `#22C55E` [accent-green] — **DYNAMIC** (green = active, red = no source)

Right group (JetBrains Mono, 9px):

- “VIEWERS” label: `rgba(255,255,255,64)`
- Viewer count: `#3B82F6` [accent-blue], weight 600 — **DYNAMIC**
- “|” separator
- “BITRATE” label: `rgba(255,255,255,64)`
- Bitrate value: `#22C55E` if ≥7000 kbps, `#F59E0B` if <7000 kbps — **DYNAMIC**

6.3 Scenes Panel

```
ScenesPanel (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px)
├── PanelHeader (QHBoxLayout, height: 32px, padding: 0 12px, border-bottom: 1px rgba(255,255,255,15))
│   ├── "SCENES" label (section header style)
│   └── ControlGroup (QHBoxLayout, spacing: 4px)
│       ├── ChevronLeft button (13×13px icon)
│       ├── ChevronRight button (13×13px icon)
│       ├── Vertical divider (1×12px, rgba(255,255,255,26))
│       ├── GridView button (13×13px icon)
│       ├── ListView button (13×13px icon)
│       └── AddScene button (Plus icon, 13×13px)
└── SceneList (QScrollArea horizontal, padding: 8px, spacing: 8px)
    └── SceneCard × N (see §6.3.1)
```

6.3.1 Scene Card Specification

```
SceneCard (QPushButton or QWidget with click handler)
Size:      110×70px (fixed)
Background: #0A0B0F
Border-radius: 6px
Overflow:  hidden (clip children)
```

State	Border	Box Shadow	Icon Color	Label Color	Label Weight
Default	2px solid rgba(255,255,255,20)	none	rgba(255,255,255,51)	rgba(255,255,255,102)	400
Hover	2px solid rgba(255,255,255,40)	none	rgba(255,255,255,102)	rgba(255,255,255,153)	400
Active/Selected	2px solid #3B82F6	0 0 12px rgba(59,130,246,77)	#3B82F6	#FFFFFF	600

Active card also shows a “LIVE” badge in top-right corner:

- Size: auto, padding: 1px 4px
- Background: #EF4444
- Font: Inter Bold, 8px, #fff, letter-spacing: 0.06em
- Border-radius: 3px
- Position: 4px from top, 4px from right (absolute)

Card internal layout (centered vertically and horizontally):

- Monitor icon: 16×16px
- Gap: 4px
- Scene name label: Inter, 10px

6.4 Audio Mixer Panel

```
AudioMixerPanel (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px, flex
├── PanelHeader (QHBoxLayout, height: 32px, padding: 0 12px, border-bottom: 1px rgba(255,255,255,15))
│   ├── "AUDIO MIXER" label
│   └── MasterControls (QHBoxLayout, spacing: 8px)
│       ├── "MASTER" label (Inter, 10px, rgba(255,255,255,102))
│       ├── Volume2 icon (12×12px, rgba(255,255,255,102))
│       ├── MasterSlider (QSlider, width: 80px, height: 4px – see §11.3)
│       ├── "-6.0 dB" label (JetBrains Mono, 10px, rgba(255,255,255,128))
│       └── Settings2 icon button (12×12px)
└── ChannelGrid (QGridLayout, 4 columns, padding: 8px, spacing: 8px)
    └── AudioChannel × 4 (see §6.4.1)
```

6.4.1 Audio Channel Card

```
AudioChannel (QWidget, background: #0E0F14, border: 1px rgba(255,255,255,15), border-radius: 6px, padding
├── ChannelHeader (QHBoxLayout, margin-bottom: 8px)
│   ├── ChannelIcon (12×12px, color: #3B82F6)
│   └── ChannelInfo (QVBoxLayout, spacing: 0)
│       ├── Channel name (Inter 600, 11px, #fff)
│       └── Channel sub-label (Inter 400, 9px, rgba(255,255,255,89))
├── VUMeter (custom QWidget, height: 16px – see §11.1)
├── VolumeSlider (QSlider, height: 3px, margin-top: 6px – see §11.3)
└── ChannelFooter (QHBoxLayout, margin-top: 6px)
    ├── dB readout (JetBrains Mono, 9px, rgba(255,255,255,102))
    └── ButtonGroup (QHBoxLayout, spacing: 4px)
        ├── SoloButton "S" (see §6.4.2)
        └── MuteButton "M" (see §6.4.2)
```

Channel data (static labels, dynamic values):

Channel	Icon	Sub-label	Level (static)	dB (static)
Mic	Mic	Shure SM7B	72%	-3.2 dB
Desktop	Monitor	System Audio	45%	-10.4 dB
Music	Music	Spotify	30%	-14.7 dB
Alert	Bell	Stream Elements	55%	-6.8 dB

Note: Channel names, sub-labels, and icon types are **DYNAMIC** from the application's audio source configuration. The level (VU meter) and dB values are **DYNAMIC** from real-time audio analysis. The static values above are prototype defaults only.

6.4.2 Solo / Mute Buttons

State	Background	Text Color	Border
Default	rgba(255,255,255,15)	rgba(255,255,255,128)	none
Hover	rgba(255,255,255,25)	rgba(255,255,255,178)	none
Active (Solo)	#F59E0B	#fff	none
Active (Mute)	#EF4444	#fff	none

Size: auto, padding: 2px 4px. Font: Inter Bold, 9px. Border-radius: 3px.

6.5 Stream Status Panel

```
StreamStatusPanel (QWidget, fixed width: 240px, background: #0E0F14, border-left: 1px rgba(255,255,255,18), padding: 12px, layout: QVBoxLayout, spacing: 12px)
├── SectionHeader "STREAM STATUS" (section header style)
├── PlatformCard (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px, padding: 10px)
│   ├── LiveDot (8×8px, #EF4444, animated pulse)
│   ├── "LIVE" (Inter 600, 12px, #EF4444)
│   └── "on YouTube" (Inter 400, 11px, rgba(255,255,255,153))
├── BitrateCard (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px, padding: 10px)
│   ├── BitrateHeader (QHBoxLayout)
│   │   └── "BITRATE" label (Inter 400, 10px, rgba(255,255,255,102), letter-spacing: 0.08em)
│   ├── BitrateValue (JetBrains Mono Bold, 13px, #3B82F6) – DYNAMIC
│   └── BitrateSparkline (custom QWidget, width: 100%, height: 40px – see §11.4)
├── StatGrid (QGridLayout, 2 columns, spacing: 8px)
│   ├── ViewersCard (see §6.5.1)
│   └── UptimeCard (see §6.5.1)
├── StreamHealthCard (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px, padding: 10px)
│   ├── HealthHeader: "STREAM HEALTH" + Activity icon
│   └── HealthRows × 3 (see §6.5.2)
└── EndStreamButton (QPushButton, full-width, height: 44px – see §6.5.3)
```

6.5.1 Stat Card (Viewers / Uptime)

```
StatCard (QWidget, background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px, padding: 10px)
├── StatHeader (QHBoxLayout)
│   ├── Icon (10×10px, color per card)
│   └── Label (Inter, 9px, rgba(255,255,255,102), letter-spacing: 0.08em, UPPERCASE)
├── Value (JetBrains Mono Bold, 15px, #fff) – DYNAMIC
└── SubLabel (Inter, 9px, accent color) – DYNAMIC
```

Card	Icon	Icon Color	Sub-label
VIEWERS	Users	#3B82F6	“Peak: 3,152” (DYNAMIC)
UPTIME	Clock	#06B6D4	“Session Live” (static label)

6.5.2 Health Row

Each row is a `QHBoxLayout` with height 20px, margin-bottom 8px:

```
[Label (50px fixed width, Inter 10px, rgba(255,255,255,128))]  
[HealthBar (flex-1, height: 4px)]  
[Value (40px fixed width, right-aligned, JetBrains Mono 10px)]
```

Row	Label	Value	Bar Color	Value Color
CPU	“CPU”	“28%” (DYNAMIC)	#22C55E if <60%, #F59E0B if <80%, #EF4444 if ≥80%	same as bar
GPU	“GPU”	“42%” (DYNAMIC)	same thresholds	same as bar
Network	“Network”	“Excellent” (DYNAMIC)	#22C55E	#22C55E

6.5.3 End Stream Button

```
QPushButton  
width:          100% (stretch)  
height:         44px  
background:     #EF4444  
color:          #FFFFFF  
font:           Space Grotesk Bold, 13px, letter-spacing: 0.04em  
border-radius:  8px  
border:         none  
box-shadow:     0 4px 16px rgba(239,68,68,77)  
icon:           Square (filled white, 14×14px) – left of text  
text:          "END STREAM"
```

State	Background	Shadow
Default	#EF4444	0 4px 16px rgba(239,68,68,77)
Hover	#DC2626 (darken 8%)	0 6px 20px rgba(239,68,68,100)
Pressed	scale(0.97) + #B91C1C	reduced shadow
Disabled	rgba(239,68,68,77)	none

7. Scene Editor Screen — Full Specification

7.1 Layout

```
SceneEditorPage (QWidget, background: #0E0F14)
├── TopBar (46px, see §5.2)
├── ContentArea (QHBoxLayout, spacing: 0)
│   ├── CanvasArea (QVBoxLayout, flex-1)
│   │   ├── CanvasWorkspace (flex-1, background: #0A0B0F, padding: 12px)
│   │   │   ├── CanvasFrame (QWidget with border: 1px rgba(255,255,255,26))
│   │   │   │   └── CanvasContent (background: #111318, see §7.2)
│   │   │   └── ToolSidebar (absolute right, vertically centered, see §7.4)
│   │   └── TransitionsBar (fixed height: 44px, border-top: 1px rgba(255,255,255,18), background: #111318)
│   └── RightPanel (fixed width: 320px, border-left: 1px rgba(255,255,255,18))
│       ├── SourcesPanel (see §7.5)
│       └── PropertiesPanel (see §7.6)
```

7.2 Canvas Content

The canvas content area fills `CanvasFrame` and contains:

- Background gradient:** `linear-gradient(135deg, #0D1117 0%, #161B22 50%, #0D1117 100%)`
- Grid overlay:** 60×60px grid, line color `QColor(255, 255, 255, 10)` (opacity ~4%), line width 0.5px
- Crosshair:** horizontal line 20×1px + vertical line 1×20px, centered, color `QColor(59, 130, 246, 64)` (rgba(59,130,246,0.25))
- Canvas metadata strip** (absolute bottom, height: 22px, background: `rgba(0,0,0,153)`, border-top: 1px `rgba(255,255,255,15)`):
 - Left: “1920×1080 · 16:9 · 60fps” (JetBrains Mono, 9px, `rgba(255,255,255,89)`)
 - Center: “SCENE 1 · 5 SOURCES” (JetBrains Mono, 9px, `rgba(59,130,246,179)`)
 - Right: “SAFE AREA: ON” (JetBrains Mono, 9px, `rgba(255,255,255,89)`)
- Safe area corner brackets:** 20×20px L-shapes, color `QColor(59, 130, 246, 102)` (rgba(59,130,246,0.4)), 8px inset from canvas edges

7.3 Source Overlay Elements (Canvas)

These are rendered as child `QWidget` s positioned absolutely within the canvas:

Selected source (Webcam) — blue selection box:

- Position: left 5%, bottom 10%, width 28%, height 38% of canvas
- Border: 2px solid `#3B82F6`
- Outer ring: 1px solid `rgba(59,130,246,77)` (box-shadow equivalent — use second border or `QPainter`)
- Background: `#1A1D2B`
- 8 resize handles: 8×8px white squares with 1px `#3B82F6` border, border-radius: 2px, positioned at 4 corners + 4 edge midpoints

Text overlay element:

- Position: top 16px, horizontally centered
- Border: 1px dashed `rgba(255,255,255,51)`
- Padding: 4px 12px
- Text: “LIVE BROADCAST”, Space Grotesk Bold, 16px, `rgba(255,255,255,179)`

Logo element:

- Position: top 12px, right 12px
- Border: 1px dashed `rgba(255,255,255,38)`
- Padding: 6px
- Inner: 48×48px rounded square, background `rgba(59,130,246,51)`, Monitor icon 18×18px, `#3B82F6`

7.4 Tool Sidebar (Vertical)

Positioned absolutely on the right edge of the canvas workspace, vertically centered. 6 buttons stacked vertically with 4px gap.

ToolButton (QPushButton, 32×32px, border-radius: 6px)

State	Background	Border	Icon Color
Default	rgba(255,255,255,15)	1px rgba(255,255,255,26)	rgba(255,255,255,128)
Hover	rgba(255,255,255,25)	1px rgba(255,255,255,40)	rgba(255,255,255,178)
Active/Selected	#3B82F6	1px #3B82F6	#FFFFFF

Tools in order (top to bottom): Move, Resize (Maximize2), Crop, Rotate (RotateCw), Align (AlignLeft), Lock

7.5 Sources Panel

SourcesPanel (QWidget, border-bottom: 1px rgba(255,255,255,18))

- SourcesHeader (QHBoxLayout, height: 32px, padding: 0 12px, border-bottom: 1px rgba(255,255,255,15))
 - "SOURCES" label (section header style)
 - ButtonGroup: Plus button + MoreHorizontal button (12×12px icons)
- SourceList (QListWidget or custom QWidget list)
 - SourceRow × N (see §7.5.1)

7.5.1 Source Row

Height: 34px. Padding: 0 12px.

State	Background	Left Border	Padding-left
Default	transparent	3px transparent	12px
Hover	rgba(255,255,255,8)	3px transparent	12px
Selected	#1A1D2B	3px solid #3B82F6	9px

Row element sequence (left to right):

- Drag handle:** :: character or 6-dot grid icon, 16×16px, rgba(255,255,255,51), cursor: grab
- Visibility toggle:** Eye / EyeOff icon, 13×13px. Visible: rgba(255,255,255,128). Hidden: rgba(255,255,255,64)
- Lock toggle:** Lock / Unlock icon, 12×12px. Locked: rgba(255,255,255,77). Unlocked: rgba(255,255,255,51)
- Type icon:** 13×13px. Selected: #3B82F6. Default: rgba(255,255,255,128)
- Source name:** Inter, 12px. Selected: #fff, weight 500. Default: rgba(255,255,255,153), weight 400
- More button:** MoreHorizontal icon, 12×12px, rgba(255,255,255,51), visible on hover only

Source list (static prototype data — **DYNAMIC** in production):

Order	Name	Icon	Visible	Locked
1	Game Capture	Monitor	true	false
2	Webcam	Camera	true	false
3	Text Overlay	Type	true	false
4	Logo Image	Image	true	true
5	Alert Overlay	Globe	false	false

7.6 Source Properties Panel

```

PropertiesPanel (QScrollArea, flex-1)
├── PropertiesHeader (QWidget, height: 32px, border-bottom: 1px rgba(255,255,255,15), padding: 0 12px)
│   └── "SOURCE PROPERTIES: [NAME]" (section header style, name in uppercase)
├── PropertiesContent (QVBoxLayout, padding: 12px, spacing: 12px)
│   ├── PositionSizeGroup (QGridLayout, 2 columns, gap: 8px)
│   │   ├── PositionGroup: POSITION label + [X input] [Y input]
│   │   └── SizeGroup: SIZE label + [W input] [H input]
│   ├── RotationOpacityGroup (QGridLayout, 2 columns, gap: 8px)
│   │   ├── RotationGroup: ROTATION label + [degrees input]
│   │   └── OpacityGroup: OPACITY label + [slider] [% label]
│   ├── ChromaKeyCard (see §7.6.1)
│   ├── FlipGroup (see §7.6.2)
│   └── CropGroup (see §7.6.3)

```

NumberInput field:

- Background: #0A0B0F
- Border: 1px rgba(255,255,255,26)
- Border-radius: 4px
- Font: JetBrains Mono, 11px, #fff
- Padding: 4px 8px
- Focus border: rgba(59,130,246,128)
- Label above: Inter, 9px, rgba(255,255,255,102), margin-bottom: 3px

7.6.1 Chroma Key Card

```

ChromaKeyCard (QWidget, background: #0E0F14, border: 1px rgba(255,255,255,18), border-radius: 4px, padding: 12px)
├── ChromaHeader (QHBoxLayout)
│   ├── "Chroma Key" (Inter 600, 11px, #fff)
│   └── ChromaToggle (Toggle widget, defaultChecked: true, color: #22C55E)
├── ChromaControls (QGridLayout, 2 columns, gap: 8px)
│   ├── ColorPicker: "Color" label + [20×20px color swatch (#22C55E)] + "#22C55E" text
│   └── Similarity: "Similarity" label + [slider, 50%] + "50%" label

```

Color swatch: 20×20px QLabel with background #22C55E, border: 1px rgba(255,255,255,51), border-radius: 3px.

7.6.2 Flip Group

Two buttons side by side:

- “↔ Flip Horizontal” and “↕ Flip Vertical”
- Style: background #0E0F14, border: 1px rgba(255,255,255,26), border-radius: 6px, padding: 6px 12px
- Font: Inter, 12px, rgba(255,255,255,153)

- Icon: FlipHorizontal2 / FlipVertical2, 12×12px, left of text

7.6.3 Crop Group

Label “CROP (px)” (section header style). 2×2 grid of NumberInputs: Top, Bottom, Left, Right, all defaulting to “0”. Below: “Reset Crop” button (full width, background rgba(255,255,255,10), border: 1px rgba(255,255,255,20), border-radius: 6px, color rgba(255,255,255,128), height: 28px).

7.7 Transitions Bar

```
TransitionsBar (QWidget, height: 44px, background: #111318, border-top: 1px rgba(255,255,255,18))
  layout: QHBoxLayout, padding: 0 12px, spacing: 16px

  └─ "TRANSITIONS" label (section header style, no-wrap)
  └─ TransitionButtons (HBoxLayout, spacing: 8px)
    └─ TransitionButton × 5: Cut, Fade, Slide, Wipe, Stinger (see §7.7.1)
  └─ DurationGroup (HBoxLayout, spacing: 8px, ml-auto)
    └─ "Duration" label (Inter, 10px, rgba(255,255,255,102))
    └─ DurationInput (QLineEdit, width: 60px, defaultValue: "300", JetBrains Mono 11px)
    └─ "ms" label (Inter, 10px, rgba(255,255,255,102))
    └─ ApplyButton ("Apply Transition", background: #3B82F6, color: #fff, Inter 600, 11px, border-radius
```

7.7.1 Transition Button

State	Background	Border	Color	Font Weight
Default	#0E0F14	1px rgba(255,255,255,20)	rgba(255,255,255,102)	400
Hover	rgba(255,255,255,10)	1px rgba(255,255,255,30)	rgba(255,255,255,153)	400
Active	rgba(59,130,246,51)	1px #3B82F6	#3B82F6	600

Size: auto, padding: 6px 12px, border-radius: 6px, font: Inter 11px.

8. Settings Screen — Full Specification

8.1 Layout

```
SettingsPage (QWidget, background: #0E0F14)
└─ TopBar (46px)
└─ ContentArea (HBoxLayout, spacing: 0, flex-1)
  └─ TabNav (QWidget, fixed width: 180px, background: #111318, border-right: 1px rgba(255,255,255,18))
    └─ TabList (QVBoxLayout, padding: 8px 0)
      └─ TabItem × 8 (see §8.2)
  └─ FormContent (QScrollArea, flex-1, padding: 24px, background: #0E0F14)
    └─ StreamTab (see §8.3) [or other tab content]
  └─ ActionBar (QWidget, fixed height: 60px, border-top: 1px rgba(255,255,255,7), background: #0D0E12)
    └─ QHBoxLayout (right-aligned)
      └─ CancelButton (ghost style, see §11.5)
      └─ SaveButton ("Save Settings", primary blue style, see §11.5)
```

8.2 Settings Tab Item

Same visual rules as sidebar nav items. Width: 180px, height: 36px, padding: 0 16px.

State	Background	Left Border	Text Color
Default	transparent	3px transparent	rgba(255,255,255,128)
Hover	rgba(255,255,255,8)	3px transparent	rgba(255,255,255,178)
Active	#1A1D2B	3px solid #3B82F6	#FFFFFF

8.3 Stream Tab Content

SERVICE section:

- Section label: “SERVICE” (section header style), margin-bottom: 10px
- Platform buttons: 4 pill buttons (Twitch, YouTube, Facebook, Custom RTMP)
 - Default: background #0E0F14, border: 1px rgba(255,255,255,20), color rgba(255,255,255,153)
 - Active: background {platformColor}22 , border: 1px {platformColor} , color {platformColor}
 - Padding: 8px 16px, border-radius: 6px, Inter 13px

FormRow pattern (used for all settings fields):

- Height: auto, min-height: 44px
- Border-bottom: 1px rgba(255,255,255,13)
- Padding: 12px 0
- Label column: fixed width 160px, Inter 500, 12px, rgba(255,255,255,153)
- Control column: flex-1

Stream Key row:

- Password input (full width minus 120px for buttons)
- “Show” button: background #0E0F14, border: 1px rgba(255,255,255,20), Inter 12px, padding: 6px 12px
- “Get Key ↗” button: background rgba(59,130,246,20), border: 1px #3B82F6, color #3B82F6, Inter 12px

ENCODER section:

- Three segmented buttons: NVENC (NVIDIA), QSV (Intel), x264 (Software)
- Active: background #3B82F6, color #fff, border: 1px #3B82F6
- Default: background #0E0F14, color rgba(255,255,255,102), border: 1px rgba(255,255,255,20)
- Padding: 8px 16px, border-radius: 6px

Encoder params grid (4 columns):

- Rate Control, Bitrate (Target), Bitrate (Min), Keyframe Interval
- Each: label above (Inter 10px, rgba(255,255,255,77)) + StyledInput below
- Bitrate inputs have “kbps” suffix label; Keyframe has “s” suffix

OUTPUT section:

- Recording Path: full-width input + “Browse” button
- Recording Format: MKV / MP4 / MOV segmented buttons (same style as encoder)
- Replay Buffer: Toggle (36×20px) + “60 seconds” label

VIDEO section:

- Canvas Resolution: QComboBox (min-width: 200px)
- Output Resolution: QComboBox (min-width: 200px)
- FPS: 30 / 60 / 120 segmented buttons

9. Chat & Audience Screen — Full Specification

9.1 Layout

```
ChatPage (QWidget, background: #0E0F14)
├── TopBar (46px)
├── ContentArea (QHBoxLayout, spacing: 0)
│   ├── LeftPanel (fixed width: 220px, border-right: 1px rgba(255,255,255,18), background: #111318)
│   │   ├── ProgramPreview (16:9 thumbnail, border-bottom: 1px rgba(255,255,255,7))
│   │   └── QuickActions (see §9.2)
│   ├── ChatCenter (flex-1, QVBoxLayout)
│   │   ├── ChatHeader (see §9.3)
│   │   ├── MessageList (QScrollArea, flex-1)
│   │   │   └── MessageRow × N (see §9.4)
│   │   ├── ChatInput (see §9.5)
│   │   └── ModToolbar (see §9.6)
│   └── RightPanel (fixed width: 260px, border-left: 1px rgba(255,255,255,18))
│       ├── ActivityFeed (see §9.7)
│       ├── PollsPanel (see §9.8)
│       └── CommandsPanel (see §9.9)
```

9.2 Quick Actions

2×2 grid of action buttons (Clip, Raid, Host, Poll) + full-width Marker button below. Each button: background rgba(255,255,255,6), border: 1px rgba(255,255,255,10), border-radius: 6px, padding: 12px 8px, icon 20×20px centered above label (Inter 11px, rgba(255,255,255,153)).

9.3 Chat Header

Height: 44px. Contains: “LIVE CHAT” label (Space Grotesk Bold, 13px, #fff) + platform color dots (T=purple, Y=red, F=blue, 8×8px circles) + viewer count badge (Users icon + count, background rgba(255,255,255,10), border-radius: 12px, padding: 4px 10px).

9.4 Message Row

Three variants:

Type	Background	Border	Notes
Regular	transparent	none	Standard chat message
Donation	rgba(245,158,11,26)	1px solid rgba(245,158,11,51)	Amber tint
Subscription	rgba(145,70,255,26)	1px solid rgba(145,70,255,51)	Purple tint

Row anatomy: [timestamp] [badge?] [username:] [message]

- Timestamp: JetBrains Mono, 10px, rgba(255,255,255,51)
- MOD badge: background rgba(34,197,94,26), border: 1px rgba(34,197,94,51), color #22C55E, Inter Bold 9px, padding: 1px 4px, border-radius: 3px
- SUB badge: background rgba(145,70,255,26), border: 1px rgba(145,70,255,51), color #9146FF, same sizing
- Username: Inter 600, 13px, color varies per user (platform-assigned)
- Message: Inter 400, 13px, rgba(255,255,255,204)

9.5 Chat Input

Height: 44px. Background: #111318. Border-top: 1px rgba(255,255,255,7). Padding: 0 12px.

- Text input: flex-1, background #0E0F14, border: 1px rgba(255,255,255,12), border-radius: 6px, Inter 12px, #fff
- Emoji button: Smile icon, 20×20px, rgba(255,255,255,77)
- Send button: 32×32px, background #3B82F6, border-radius: 6px, Send icon 14×14px white

9.6 Mod Toolbar

Height: 36px. Row of icon-only buttons (Shield, Ban, Clock, Users, MessageSquare, Trash2), each 28×28px, background rgba(255,255,255,6), border-radius: 4px, icon 14×14px rgba(255,255,255,77).

9.7 Activity Feed

Section header “ACTIVITY FEED”. Each event row: icon (16×16px, colored) + text (Inter 12px). Event types: Follow (Heart, #EF4444), Sub (Star, #9146FF), Donation (DollarSign, #F59E0B), Raid (Zap, #06B6D4). Username in event: colored bold. Timestamp: JetBrains Mono 9px, rgba(255,255,255,51).

9.8 Polls & Predictions

Section header “POLLS & PREDICTIONS” + “ACTIVE” badge (background rgba(34,197,94,26), border: 1px rgba(34,197,94,51), color #22C55E, 9px Inter Bold). Poll question: Inter 600, 13px, #fff. Each option: label + progress bar (4px height, #3B82F6) + percentage. Total votes + countdown timer in JetBrains Mono 10px.

9.9 Chat Commands

Section header “CHAT COMMANDS”. 2-column grid of command cards. Each card: background rgba(255,255,255,6), border: 1px rgba(255,255,255,10), border-radius: 6px, padding: 8px 10px. Command: Inter 600, 12px, #3B82F6. Description: Inter 400, 10px, rgba(255,255,255,77).

10. Analytics Screen — Full Specification

10.1 Layout

```
AnalyticsPage (QWidget, background: #0E0F14)
├── TopBar (46px)
├── ScrollContent (QScrollArea)
│   ├── QVBoxLayout (padding: 12px, spacing: 12px)
│   │   ├── KPIGrid (QGridLayout, 4 columns, spacing: 12px)
│   │   │   └── KPICard × 4 (see §10.2)
│   │   ├── ViewerChart (see §10.3)
│   │   ├── MiddleRow (QHBoxLayout, spacing: 12px)
│   │   │   ├── StreamHealthPanel (flex-1)
│   │   │   └── AudienceBreakdownPanel (flex-1)
│   │   └── SessionsTable (see §10.4)
```

10.2 KPI Card

```
KPICard (QWidget, background: #111318, border: 1px solid {accentColor}48, border-radius: 8px, padding: 12px)
├── CardHeader (QHBoxLayout)
│   ├── Label (Inter 600, 9px, rgba(255,255,255,102), UPPERCASE, letter-spacing: 0.1em)
│   └── Icon (13×13px, color: accentColor)
├── Value (Space Grotesk Bold, 20px, #fff) – DYNAMIC
└── Delta (Inter, 11px, color: green if positive, red if negative) – DYNAMIC
```

Card	Label	Icon	Accent Color
PEAK VIEWERS	Users	#3B82F6	#3B82F6
AVG. WATCH TIME	Clock	#06B6D4	#06B6D4
NEW FOLLOWERS	UserPlus	#22C55E	#22C55E
TOTAL REVENUE	DollarSign	#F59E0B	#F59E0B

10.3 Viewer Chart

Full-width panel (background: #111318, border: 1px rgba(255,255,255,20), border-radius: 8px, padding: 12px). Chart height: 140px. Use `QCustomPlot` or `QtCharts::QChart` .

- Line color: #3B82F6, width: 1.5px
- Fill: gradient from rgba(59,130,246,102) at top to rgba(59,130,246,0) at bottom
- Grid lines: rgba(255,255,255,13), 1px dashed
- Axis labels: JetBrains Mono, 9px, rgba(255,255,255,77)
- Time range selector: 6 pill buttons (5m, 15m, 1h, 3h, 6h, 12h), active: background rgba(59,130,246,51), border: 1px #3B82F6, color #3B82F6
- LIVE DATA badge: red pulse dot + “LIVE DATA” text, same as LIVE badge style
- Instrumentation strip above chart: PEAK / AVG / CURRENT values in JetBrains Mono 10px

10.4 Sessions Table

Panel with “RECENT STREAM SESSIONS” header. Standard HTML-style table rendered via `QTableWidget` .

Column	Alignment	Color	Notes
Date	Left	rgba(255,255,255,153)	Inter 12px
Duration	Left	rgba(255,255,255,153)	Inter 12px
Peak Viewers	Left	#3B82F6	JetBrains Mono 12px Bold
Avg Viewers	Left	rgba(255,255,255,153)	Inter 12px
Revenue	Left	#22C55E	JetBrains Mono 12px Bold
Quality Score	Center	Badge widget	Excellent=green, Good=blue, Fair=amber

Table styling: background #111318, header background #0E0F14, row separator: 1px rgba(255,255,255,7), alternating row: rgba(255,255,255,4), hover row: rgba(255,255,255,8).

11. Reusable Component Specifications

11.1 VU Meter (Audio Level Indicator)

Implement as a custom `QWidget` subclass with `paintEvent` .

```
VUMeter
  Container size:  variable width × 16px height
  Bar count:      20
  Bar width:      3px each
  Bar gap:        1px
  Bar border-radius: 1px
  Bar heights:    4px + (index % 3)px  [4, 5, 6, 4, 5, 6, ... pattern]
  Alignment:      bottom-aligned within container
```

Color thresholds (based on bar index, 0–19):

Bar index	Active color	Inactive color
0–13	#22C55E	rgba(255,255,255,26)
14–16	#F59E0B	rgba(255,255,255,26)
17–19	#EF4444	rgba(255,255,255,26)

A bar is “active” when $(\text{barIndex} / \text{totalBars}) * 100 < \text{currentLevel}$. `currentLevel` is 0–100 (percentage).

Update rate: 50ms timer (20fps minimum, 60fps preferred). Transition: `background` color change, 50ms.

In `paintEvent`:

```
void VUMeter::paintEvent(QPaintEvent*) {
    QPainter p(this);
    p.setRenderHint(QPainter::Antialiasing, false);
    int barW = 3, gap = 1, totalW = barW + gap;
    for (int i = 0; i < 20; i++) {
        int barH = 4 + (i % 3);
        float threshold = (float)i / 20.0f * 100.0f;
        bool active = threshold < m_level;
        QColor color = active
            ? (i < 14 ? QColor(34,197,94) : i < 17 ? QColor(245,158,11) : QColor(239,68,68))
            : QColor(255,255,255,26);
        p.fillRect(i * totalW, height() - barH, barW, barH,
            QBrush(color, Qt::SolidPattern));
    }
}
```

11.2 Toggle Switch

Implement as a custom `QWidget` subclass or `QPushButton` with custom `paintEvent`.

Settings screen variant (larger):

- Container: 36×20px, border-radius: 10px
- Thumb: 16×16px circle, border-radius: 8px, color #fff
- Thumb position ON: left = 18px (2px from right edge)
- Thumb position OFF: left = 2px
- Background ON: #3B82F6
- Background OFF: rgba(255,255,255,38)
- Transition: thumb left position, 200ms ease

Scene Editor variant (smaller, chroma key):

- Container: 32×18px, border-radius: 9px
- Thumb: 14×14px circle
- Thumb position ON: left = 16px
- Thumb position OFF: left = 2px
- Background ON: #22C55E (green for chroma key active)
- Background OFF: rgba(255,255,255,38)

11.3 Slider (Volume / Opacity)

Implement via `QSlider` with custom `QSS`:

```

QSlider::groove:horizontal {
    height: 4px;
    background: rgba(255,255,255,20);
    border-radius: 2px;
}
QSlider::sub-page:horizontal {
    background: #3B82F6;
    border-radius: 2px;
}
QSlider::handle:horizontal {
    width: 12px;
    height: 12px;
    background: #fff;
    border-radius: 6px;
    margin: -4px 0;
    border: 2px solid #3B82F6;
}
QSlider::handle:horizontal:hover {
    background: #3B82F6;
}

```

For the 3px-height variant (channel volume, opacity, similarity): set groove height to 3px, handle margin to -4.5px 0.

11.4 Bitrate Sparkline

Custom `QWidget` subclass. Stores a rolling buffer of 30 bitrate samples. On each `paintEvent`:

1. Normalize values to 0–1 range based on max in buffer
2. Draw filled area: `QPainterPath` from left to right, closed at bottom
 - Fill: `QLinearGradient` from `QColor(59,130,246,102)` at top to `QColor(59,130,246,0)` at bottom
3. Draw line: `QPen(QColor(59,130,246), 1.5f)`, `Qt::SolidLine`, `Qt::RoundCap`
4. Update via 1-second `QTimer`, append new sample, remove oldest

Widget size: full panel width × 40px.

11.5 Button Variants

Primary (Blue):

```

QPushButton[variant="primary"] {
    background: #3B82F6;
    color: #ffffff;
    border: none;
    border-radius: 6px;
    padding: 8px 16px;
    font-family: "Inter";
    font-size: 12px;
    font-weight: 600;
}
QPushButton[variant="primary"]:hover { background: #2563EB; }
QPushButton[variant="primary"]:pressed { background: #1D4ED8; }
QPushButton[variant="primary"]:disabled { background: rgba(59,130,246,77); color: rgba(255,255,255,128); }

```

Ghost (Cancel):

```

QPushButton[variant="ghost"] {
    background: transparent;
    color: rgba(255,255,255,153);
    border: 1px solid rgba(255,255,255,20);
    border-radius: 6px;
    padding: 8px 16px;
    font-family: "Inter";
    font-size: 12px;
}
QPushButton[variant="ghost"]:hover { background: rgba(255,255,255,8); color: rgba(255,255,255,204); }
QPushButton[variant="ghost"]:pressed { background: rgba(255,255,255,13); }

```

Danger (End Stream):

```

QPushButton[variant="danger"] {
    background: #EF4444;
    color: #ffffff;
    border: none;
    border-radius: 8px;
    padding: 10px 16px;
    font-family: "Space Grotesk";
    font-size: 13px;
    font-weight: 700;
    letter-spacing: 0.04em;
}
QPushButton[variant="danger"]:hover { background: #DC2626; }
QPushButton[variant="danger"]:pressed { background: #B91C1C; }

```

Icon-only toolbar button:

```

QPushButton[variant="tool"] {
    background: rgba(255,255,255,15);
    border: 1px solid rgba(255,255,255,26);
    border-radius: 6px;
    padding: 0;
    min-width: 32px;
    min-height: 32px;
    max-width: 32px;
    max-height: 32px;
}
QPushButton[variant="tool"]:hover { background: rgba(255,255,255,25); }
QPushButton[variant="tool"]:checked { background: #3B82F6; border-color: #3B82F6; }

```

11.6 Health / Progress Bar

Custom `QWidget` or `QProgressBar` with QSS:

```

QProgressBar {
    background: rgba(255,255,255,20);
    border-radius: 2px;
    border: none;
    max-height: 4px;
    min-height: 4px;
    text-visible: false;
}
QProgressBar::chunk { border-radius: 2px; }
QProgressBar[health="good"]::chunk { background: #22C55E; }
QProgressBar[health="warn"]::chunk { background: #F59E0B; }
QProgressBar[health="bad"]::chunk { background: #EF4444; }

```


11.7 LIVE Pulse Animation

The red dot in LIVE badges and the sidebar footer dot must animate. Use `QPropertyAnimation` on a custom property or `QTimer` - driven opacity oscillation:

```
// Pseudocode - drive with QTimer at 50ms intervals
float t = (elapsed_ms % 2000) / 2000.0f;
float opacity = 0.5f + 0.5f * cos(t * 2 * M_PI); // 1.0 → 0.5 → 1.0 over 2 seconds
dot->setOpacity(opacity);
// Also scale: 1.0 → 1.3 → 1.0
```

12. All Widget State Specifications

12.1 Complete State Matrix

Widget	Default	Hover	Active/Selected	Pressed	Disabled	LIV
Nav item	transparent bg, 50% text	rgba(255,255,255,5%) bg	#1A1D2B bg + 3px blue border	rgba(255,255,255,8%)	30% opacity	—
Scene card	rgba(255,255,255,8%) border	rgba(255,255,255,16%) border	2px #3B82F6 border + blue glow	scale(0.98)	40% opacity	rec LIVE badge
Source row	transparent	rgba(255,255,255,3%)	#1A1D2B + 3px blue border	—	40% opacity	—
Tool button	rgba(255,255,255,6%)	rgba(255,255,255,10%)	#3B82F6 bg + white icon	scale(0.97)	40% opacity	—
Transition btn	#0E0F14	rgba(255,255,255,4%)	rgba(blue,20%) + blue border	—	40% opacity	—
Platform btn	#0E0F14	rgba(255,255,255,4%)	platform-color bg/border	—	40% opacity	—
Toggle (off)	rgba(255,255,255,15%)	rgba(255,255,255,20%)	—	—	40% opacity	—
Toggle (on)	#3B82F6	#2563EB	—	—	rgba(blue,40%)	—
Primary button	#3B82F6	#2563EB	—	#1D4ED8 + scale(0.97)	rgba(blue,30%)	—
Danger button	#EF4444	#DC2626	—	#B91C1C + scale(0.97)	rgba(red,30%)	—
Input field	#0E0F14 + rgba(white,12%) border	rgba(white,18%) border	—	—	40% opacity	—
Health bar	—	—	—	—	—	—
VU bar	rgba(white,10%)	—	—	—	—	—
LIVE badge	red bg/border + pulse	—	—	—	hidden	show
SIG OK dot	green	—	—	—	—	—
Bitrate value	#3B82F6	—	—	—	—	—

12.2 LIVE State (Streaming Active)

When the application is in LIVE state:

- Top bar LIVE badge: visible, red, pulsing
- Sidebar footer: red dot + “LIVE · HH:MM:SS” (timecode updating)
- Active scene card: red “LIVE” badge in top-right corner
- Stream Status panel: all values updating in real time

- End Stream button: fully opaque and enabled
- Instrumentation strip: timecode, viewer count, bitrate all live

12.3 Offline State (Not Streaming)

When the application is offline (not streaming):

- Top bar LIVE badge: hidden or replaced with “OFFLINE” in rgba(255,255,255,40%)
- Sidebar footer: gray dot + “OFFLINE”
- Active scene card: no LIVE badge
- Stream Status panel: all values show “—” or last known values grayed out
- End Stream button: replaced with “GO LIVE” button (same size, #22C55E background)
- Bitrate sparkline: flat line or empty

12.4 Warning State

Triggered when bitrate drops below 4000 kbps or CPU exceeds 80%:

- Affected health bar: turns #F59E0B (amber)
- Affected value label: turns #F59E0B
- SIG OK indicator: amber dot + “DEGRADED” text
- Bitrate value in top bar: turns #F59E0B

12.5 Error/Critical State

Triggered when bitrate drops below 2000 kbps, dropped frames exceed 5%, or network disconnects:

- Affected health bar: turns #EF4444 (red)
- SIG OK indicator: red dot + “NO SIGNAL” or “DROPPED FRAMES”
- Top bar LIVE badge: flashes rapidly (200ms pulse instead of 2000ms)
- Alert toast: appears at top-right, background #EF4444, white text

13. SVG Icon Inventory

All icons are from the **Lucide** icon set (v0.453.0). All icons use `stroke="currentColor"`, `stroke-width="2"`, `stroke-linecap="round"`, `stroke-linejoin="round"`, `fill="none"`, `viewBox="0 0 24 24"`.

To use in Qt: load SVG via `QSvgRenderer` or `QIcon(QPixmap::fromImage(...))`. Set color by replacing `currentColor` with the desired hex before loading, or use `QSvgRenderer` with a custom color filter.

13.1 Icon Usage Map

Icon Name	File	Used In	Size(s)
radio	radio.svg	Sidebar logo	15px
layout-dashboard	layout-dashboard.svg	Sidebar nav	16px
layers	layers.svg	Sidebar nav	16px
message-square	message-square.svg	Sidebar nav	16px
bar-chart-2	bar-chart-2.svg	Sidebar nav	16px
settings	settings.svg	Sidebar nav	16px
wifi	wifi.svg	Sidebar footer	13px
wifi-off	wifi-off.svg	Sidebar footer (offline)	13px
monitor	monitor.svg	Scene card placeholder, canvas placeholder, audio channel	16–32px
camera	camera.svg	Webcam source icon, canvas	13–20px
type	type.svg	Text Overlay source icon	13px
image	image.svg	Logo Image source icon	13px
globe	globe.svg	Alert Overlay source icon	13px
eye	eye.svg	Source visibility (visible)	13px
eye-off	eye-off.svg	Source visibility (hidden)	13px
lock	lock.svg	Source locked	12px
unlock	unlock.svg	Source unlocked	12px
more-horizontal	more-horizontal.svg	Source row overflow menu	12px
plus	plus.svg	Add scene, add source	13px
move	move.svg	Canvas tool: Move	14px
maximize-2	maximize-2.svg	Canvas tool: Resize	14px
crop	crop.svg	Canvas tool: Crop	14px
rotate-cw	rotate-cw.svg	Canvas tool: Rotate	14px
align-left	align-left.svg	Canvas tool: Align	14px
flip-horizontal-2	flip-horizontal-2.svg	Flip Horizontal button	12px
flip-vertical-2	flip-vertical-2.svg	Flip Vertical button	12px
volume-2	volume-2.svg	Master volume icon	12px
users	users.svg	Viewers stat, KPI card	10–13px
clock	clock.svg	Uptime stat, KPI card	10–11px
zap	zap.svg	Raid event icon	16px
heart	heart.svg	Follow event icon	16px
star	star.svg	Sub event icon	16px
dollar-sign	dollar-sign.svg	Donation event, revenue KPI	13–16px

Icon Name	File	Used In	Size(s)
user-plus	user-plus.svg	New followers KPI	13px
shield	shield.svg	Mod tool	14px
ban	ban.svg	Ban tool	14px
trash-2	trash-2.svg	Clear chat tool	14px
send	send.svg	Chat send button	14px
smile	smile.svg	Chat emoji button	20px
flag	flag.svg	Chat report	14px
refresh-cw	refresh-cw.svg	Analytics refresh	11–12px
download	download.svg	Analytics export	10–11px
alert-circle	alert-circle.svg	Error/warning state	14px
alert-triangle	alert-triangle.svg	Warning state	14px
check	check.svg	Success state	14px
chevron-down	chevron-down.svg	Select dropdowns	12px
chevron-right	chevron-right.svg	Breadcrumb separators	12px
arrow-left	arrow-left.svg	Back navigation	16px
arrow-right	arrow-right.svg	Scene scroll	13px
sliders	sliders.svg	Advanced settings tab	16px
keyboard	keyboard.svg	Hotkeys settings tab	16px
plug	plug.svg	Plugins settings tab	16px
scissors	scissors.svg	Clip action	20px
rotate-ccw	rotate-ccw.svg	Undo	14px
camera	camera.svg	Camera action	20px

13.2 Loading SVG Icons in Qt

```
// Method 1: QIcon from SVG file (recommended for scalable icons)
QIcon icon(":/icons/radio.svg");
button->setIcon(icon);
button->setIconSize(QSize(15, 15));

// Method 2: Colorize SVG at runtime (for dynamic color changes)
QSvgRenderer renderer(QString(":/icons/radio.svg"));
QPixmap pixmap(16, 16);
pixmap.fill(Qt::transparent);
QPainter painter(&pixmap);
renderer.render(&painter);
// Apply color via QPainter::CompositionMode_SourceIn
painter.setCompositionMode(QPainter::CompositionMode_SourceIn);
painter.fillRect(pixmap.rect(), QColor(59, 130, 246)); // accent-blue
painter.end();
QIcon coloredIcon(pixmap);
```

14. Static vs. Dynamic Data Guide

This section identifies every value in the prototype and classifies it as either **STATIC** (a hardcoded design example) or **DYNAMIC** (must be driven by live application data).

14.1 Dashboard Screen

Element	Value in Prototype	Classification	Data Source
Platform abbreviation “YT”	“YT”	DYNAMIC	Active streaming service config
Timecode in top bar	“01:23:49”	DYNAMIC	<code>QEelapsedTimer</code> since stream start
Resolution “1920×1080”	“1920×1080”	DYNAMIC	Encoder output resolution setting
FPS “60fps”	“60fps”	DYNAMIC	Encoder FPS setting
Codec “H.264”	“H.264”	DYNAMIC	Encoder codec setting
SIG OK indicator	green dot + “SIG OK”	DYNAMIC	Network/encoder health check
Profile name “Default Profile”	“Default Profile”	DYNAMIC	Active profile name
Canvas content	black + grid + placeholder	DYNAMIC	Live video frame from encoder
TC in instrumentation strip	“01:23:49:02”	DYNAMIC	Timecode + frame counter
Scene name in strip	“Scene 1”	DYNAMIC	Active scene name
SRC ACTIVE in strip	“SRC ACTIVE”	DYNAMIC	Whether active scene has sources
Viewer count in strip	“2,843”	DYNAMIC	Platform API viewer count
Bitrate in strip	“8300 kbps”	DYNAMIC	Encoder output bitrate
Scene list names	“Scene 1” ... “End Screen”	DYNAMIC	User-configured scene list
Scene card thumbnails	black placeholder	DYNAMIC	Scene preview frames
Active scene indicator	“Scene 1”	DYNAMIC	Currently active scene
Audio channel names	“Mic”, “Desktop”, etc.	DYNAMIC	Configured audio sources
Audio channel sub-labels	“Shure SM7B”, etc.	DYNAMIC	Audio device names
VU meter levels	72%, 45%, 30%, 55%	DYNAMIC	Real-time audio level analysis
dB readouts	“-3.2 dB”, etc.	DYNAMIC	Real-time audio level in dB
Master volume	70% / “-6.0 dB”	DYNAMIC	Master volume setting
LIVE platform label	“on YouTube”	DYNAMIC	Active streaming platform
Bitrate value	“8,347 kbps”	DYNAMIC	Encoder output bitrate
Bitrate sparkline	animated waveform	DYNAMIC	Rolling 30-sample bitrate history
Viewer count	“2,842”	DYNAMIC	Platform API
Peak viewers	“Peak: 3,152”	DYNAMIC	Session max viewer count
Uptime	“01:23:49”	DYNAMIC	<code>QEelapsedTimer</code> since stream start
CPU %	“28%”	DYNAMIC	System CPU usage (e.g., <code>GetSystemTimes</code>)
GPU %	“42%”	DYNAMIC	GPU usage (NVAPI / D3DKMT)
Network health	“Excellent”	DYNAMIC	Packet loss / RTT to ingest server

14.2 Scene Editor Screen

Element	Value in Prototype	Classification	Data Source
Project name chip	"My Broadcast Project"	DYNAMIC	Active project/profile name
LIVE timecode	"00:00:00"	DYNAMIC	Stream elapsed time
REC timecode	"00:00:00"	DYNAMIC	Recording elapsed time
CPU %	"2.1%"	DYNAMIC	System CPU usage
FPS	"60.00"	DYNAMIC	Encoder output FPS
KB/s	"5120"	DYNAMIC	Encoder output bitrate
Canvas background	gradient placeholder	DYNAMIC	Composited scene preview
Canvas metadata	"1920×1080 · 16:9 · 60fps"	DYNAMIC	Canvas resolution settings
Scene name in strip	"SCENE 1"	DYNAMIC	Active scene name
Source count	"5 SOURCES"	DYNAMIC	Count of sources in active scene
SAFE AREA toggle	"ON"	DYNAMIC	User setting
Source list	5 sources	DYNAMIC	Scene source list
Source visibility	eye/eye-off state	DYNAMIC	Per-source visibility setting
Source lock state	lock/unlock icon	DYNAMIC	Per-source lock setting
Selected source	"Webcam"	DYNAMIC	User selection
Source position X/Y	"80.0", "720.0"	DYNAMIC	Source transform data
Source size W/H	"640.0", "360.0"	DYNAMIC	Source transform data
Rotation	"0.0"	DYNAMIC	Source transform data
Opacity	"100%"	DYNAMIC	Source opacity setting
Chroma key enabled	true	DYNAMIC	Source filter setting
Chroma key color	"#22C55E"	DYNAMIC	Source filter setting
Similarity	"50%"	DYNAMIC	Source filter setting
Crop values	"0" (all sides)	DYNAMIC	Source crop setting
Active transition	"Cut"	DYNAMIC	User-selected transition
Duration	"300" ms	DYNAMIC	Transition duration setting
Text overlay content	"LIVE BROADCAST"	DYNAMIC	Text source content

14.3 Settings Screen

Element	Value in Prototype	Classification	Data Source
Active platform	"YouTube"	DYNAMIC	Stream service config
Stream key	"....."	DYNAMIC	Stored stream key (masked)
Server	"Auto-select (recommended)"	DYNAMIC	Selected ingest server
Active encoder	"NVENC (NVIDIA)"	DYNAMIC	Encoder config
Rate Control	"CBR"	DYNAMIC	Encoder config
Bitrate Target	"6000"	DYNAMIC	Encoder config
Bitrate Min	"4500"	DYNAMIC	Encoder config
Keyframe Interval	"2"	DYNAMIC	Encoder config
Recording Path	"D:\StreamPro\Recordings"	DYNAMIC	User-configured path
Recording Format	"MKV"	DYNAMIC	User config
Replay Buffer	enabled + "60 seconds"	DYNAMIC	User config
Canvas Resolution	"1920×1080 (16:9)"	DYNAMIC	Video config
Output Resolution	"1920×1080 (16:9)"	DYNAMIC	Video config
FPS	"60"	DYNAMIC	Video config
LIVE indicator in top bar	red dot	DYNAMIC	Stream state
"Changes apply on next stream"	static notice	STATIC	UI copy

14.4 Chat Screen

Element	Classification	Notes
Chat messages	DYNAMIC	Platform chat API (Twitch IRC / YouTube Live Chat API)
Viewer count badge	DYNAMIC	Platform API
Activity feed events	DYNAMIC	Platform events API
Poll question and results	DYNAMIC	Platform polls API
Poll countdown	DYNAMIC	Timer from poll end time
Chat command list	DYNAMIC	User-configured commands
Program preview thumbnail	DYNAMIC	Live video frame
Timecode on preview	DYNAMIC	Stream elapsed time

14.5 Analytics Screen

Element	Classification	Notes
All KPI values	DYNAMIC	Platform API + local session data
Delta percentages	DYNAMIC	Calculated vs. previous session
Viewer count chart	DYNAMIC	Time-series from platform API
Bitrate / CPU / GPU / FPS lines	DYNAMIC	Local encoder telemetry
Dropped frames %	DYNAMIC	Encoder dropped frame counter
Audience breakdown (Twitch 54% etc.)	DYNAMIC	Platform API
Top countries	DYNAMIC	Platform API
Recent sessions table	DYNAMIC	Local session history database
Time range selector state	DYNAMIC	User selection

15. Qt Widget & Layout Mapping

15.1 Widget Selection Guide

UI Element	Recommended Qt Widget	Notes
Application window	<code>QMainWindow</code>	Use <code>setCentralWidget()</code> for the main split
Sidebar	<code>QWidget</code> with <code>QVBoxLayout</code>	Fixed width via <code>setFixedWidth(220)</code>
Page navigation	<code>QStackedWidget</code>	Switch pages via <code>setCurrentIndex()</code>
Nav items	<code>QPushButton</code> or <code>QLabel</code> with mouse events	Custom <code>paintEvent</code> for left-border active state
Top bar	<code>QWidget</code> with <code>QHBoxLayout</code>	Fixed height via <code>setFixedHeight(46)</code>
Panel/card	<code>QFrame</code> with <code>QVBoxLayout</code>	Set <code>setFrameShape(QFrame::NoFrame)</code> , use QSS for border
Section header	<code>QLabel</code>	Uppercase via <code>text().toUpper()</code> or QSS <code>text-transform</code>
Scene card	<code>QPushButton</code>	Custom <code>paintEvent</code> for LIVE badge overlay
Scene list	<code>QScrollArea</code> + <code>QWidget</code> + <code>QHBoxLayout</code>	Horizontal scroll, <code>setHorizontalScrollBarPolicy(Qt::ScrollBarAsNeeded)</code>
Audio channel	<code>QWidget</code> with <code>QVBoxLayout</code>	Custom card
VU meter	<code>QWidget</code> subclass	Custom <code>paintEvent</code> (see §11.1)
Volume slider	<code>QSlider</code>	Custom QSS (see §11.3)
Toggle switch	<code>QWidget</code> subclass	Custom <code>paintEvent</code> + <code>mousePressEvent</code> (see §11.2)
Bitrate sparkline	<code>QWidget</code> subclass	Custom <code>paintEvent</code> (see §11.4)
Health bar	<code>QProgressBar</code>	Custom QSS (see §11.6)
Text input	<code>QLineEdit</code>	Custom QSS for border/background
Password input	<code>QLineEdit</code> with <code>setEchoMode(QLineEdit::Password)</code>	
Dropdown select	<code>QComboBox</code>	Custom QSS for dark styling
Settings form	<code>QWidget</code> + <code>QFormLayout</code> or <code>QGridLayout</code>	Fixed label column width
Chat messages	<code>QScrollArea</code> + <code>QWidget</code> + <code>QVBoxLayout</code>	Auto-scroll to bottom on new message
Chat input	<code>QLineEdit</code>	
Analytics chart	<code>QCustomPlot</code> (third-party) or <code>QtCharts::QChartView</code>	<code>QCustomPlot</code> recommended for performance
Sessions table	<code>QTableWidget</code>	Custom item delegates for colored cells
Color swatch	<code>QLabel</code>	Set background via QSS
Donut chart	<code>QWidget</code> subclass	Custom <code>paintEvent</code> with <code>QPainter::drawArc</code>
Tooltip	<code>QToolTip</code>	Style via <code>QToolTip</code> QSS
Toast/notification	<code>QFrame</code> (overlay)	Position via <code>QPropertyAnimation</code>

UI Element	Recommended Qt Widget	Notes
Separator (vertical)	<code>QFrame</code> with <code>setFrameShape(QFrame::VLine)</code>	
Separator (horizontal)	<code>QFrame</code> with <code>setFrameShape(QFrame::HLine)</code>	
Scroll area	<code>QScrollArea</code>	Hide scrollbar via QSS when not needed
Icon button	<code>QPushButton</code> with <code>setIcon()</code>	<code>setFlat(true)</code> + custom QSS
Badge/chip	<code>QLabel</code>	Rounded via QSS <code>border-radius</code>

15.2 Layout Hierarchy (Dashboard)

```
// Pseudocode structure
QWidget* dashPage = new QWidget;
QVBoxLayout* dashLayout = new QVBoxLayout(dashPage);
dashLayout->setSpacing(0);
dashLayout->setContentsMargins(0,0,0,0);
dashLayout->addWidget(topBar); // fixed 46px

QHBoxLayout* content = new QHBoxLayout;
content->setSpacing(0);
content->setContentsMargins(0,0,0,0);

// Center column
QWidget* center = new QWidget;
QVBoxLayout* centerLayout = new QVBoxLayout(center);
centerLayout->setSpacing(12);
centerLayout->setContentsMargins(12,12,12,12);
centerLayout->addWidget(programOutputPanel); // aspect-ratio constrained
centerLayout->addWidget(scenesPanel); // fixed height ~100px
centerLayout->addWidget(audioMixerPanel, 1); // flex-1

// Right panel
QWidget* rightPanel = new QWidget;
rightPanel->setFixedWidth(240);
// ... add stream status widgets

content->addWidget(center, 1);
content->addWidget(rightPanel);
dashLayout->addLayout(content, 1);
```

16. Complete QSS Stylesheet

Apply this stylesheet to `QApplication::setStyleSheet()` at startup. Use `setProperty()` with variant attributes for state-specific rules.

```

/* =====
NEXUS BROADCAST - Global QSS Stylesheet
Target: Qt 6 Widgets
Version: 1.0
===== */

/* --- Reset & Base --- */
* {
    outline: none;
    box-sizing: border-box;
}

QWidget {
    background-color: #0E0F14;
    color: #FFFFFF;
    font-family: "Inter";
    font-size: 12px;
    border: none;
}

/* --- Scrollbars --- */
QScrollBar:vertical {
    background: transparent;
    width: 6px;
    margin: 0;
}
QScrollBar::handle:vertical {
    background: rgba(255,255,255,40);
    border-radius: 3px;
    min-height: 20px;
}
QScrollBar::handle:vertical:hover { background: rgba(255,255,255,70); }
QScrollBar::add-line:vertical, QScrollBar::sub-line:vertical { height: 0; }
QScrollBar::add-page:vertical, QScrollBar::sub-page:vertical { background: transparent; }

QScrollBar:horizontal {
    background: transparent;
    height: 6px;
    margin: 0;
}
QScrollBar::handle:horizontal {
    background: rgba(255,255,255,40);
    border-radius: 3px;
    min-width: 20px;
}
QScrollBar::handle:horizontal:hover { background: rgba(255,255,255,70); }
QScrollBar::add-line:horizontal, QScrollBar::sub-line:horizontal { width: 0; }

/* --- QLineEdit (Text Input) --- */
QLineEdit {
    background: #0E0F14;
    border: 1px solid rgba(255,255,255,31);
    border-radius: 6px;
    color: #FFFFFF;
    font-family: "Inter";
    font-size: 12px;
    padding: 6px 12px;
    selection-background-color: rgba(59,130,246,100);
}
QLineEdit:focus {
    border: 1px solid rgba(59,130,246,128);
}
QLineEdit:disabled {

```

```

        color: rgba(255,255,255,51);
        border-color: rgba(255,255,255,13);
    }
    QLineEdit[monospace="true"] {
        font-family: "JetBrains Mono";
    }

    /* --- QComboBox (Select / Dropdown) --- */
    QComboBox {
        background: #0E0F14;
        border: 1px solid rgba(255,255,255,31);
        border-radius: 6px;
        color: #FFFFFF;
        font-family: "Inter";
        font-size: 12px;
        padding: 6px 12px;
        min-width: 200px;
    }
    QComboBox:focus { border-color: rgba(59,130,246,128); }
    QComboBox::drop-down {
        border: none;
        width: 24px;
    }
    QComboBox::down-arrow {
        image: url(/icons/chevron-down.svg);
        width: 12px;
        height: 12px;
    }
    QComboBox QAbstractItemView {
        background: #16181F;
        border: 1px solid rgba(255,255,255,20);
        border-radius: 6px;
        color: #FFFFFF;
        selection-background-color: rgba(59,130,246,51);
        padding: 4px;
    }

    /* --- QPushButton (Primary) --- */
    QPushButton[variant="primary"] {
        background: #3B82F6;
        color: #FFFFFF;
        border: none;
        border-radius: 6px;
        padding: 8px 16px;
        font-family: "Inter";
        font-size: 12px;
        font-weight: 600;
    }
    QPushButton[variant="primary"]:hover { background: #2563EB; }
    QPushButton[variant="primary"]:pressed { background: #1D4ED8; }
    QPushButton[variant="primary"]:disabled {
        background: rgba(59,130,246,77);
        color: rgba(255,255,255,128);
    }

    /* --- QPushButton (Ghost) --- */
    QPushButton[variant="ghost"] {
        background: transparent;
        color: rgba(255,255,255,153);
        border: 1px solid rgba(255,255,255,51);
        border-radius: 6px;
        padding: 8px 16px;
        font-family: "Inter";
        font-size: 12px;
    }

```

```

}
QPushButton[variant="ghost"]:hover {
    background: rgba(255,255,255,20);
    color: rgba(255,255,255,204);
}
QPushButton[variant="ghost"]:pressed { background: rgba(255,255,255,33); }

/* --- QPushButton (Danger) --- */
QPushButton[variant="danger"] {
    background: #EF4444;
    color: #FFFFFF;
    border: none;
    border-radius: 8px;
    padding: 10px 16px;
    font-family: "Space Grotesk";
    font-size: 13px;
    font-weight: 700;
}
QPushButton[variant="danger"]:hover { background: #DC2626; }
QPushButton[variant="danger"]:pressed { background: #B91C1C; }
QPushButton[variant="danger"]:disabled { background: rgba(239,68,68,77); }

/* --- QPushButton (Tool) --- */
QPushButton[variant="tool"] {
    background: rgba(255,255,255,15);
    border: 1px solid rgba(255,255,255,26);
    border-radius: 6px;
    min-width: 32px;
    max-width: 32px;
    min-height: 32px;
    max-height: 32px;
    padding: 0;
}
QPushButton[variant="tool"]:hover { background: rgba(255,255,255,25); }
QPushButton[variant="tool"]:checked {
    background: #3B82F6;
    border-color: #3B82F6;
}

/* --- QPushButton (Icon-only small) --- */
QPushButton[variant="icon"] {
    background: transparent;
    border: none;
    padding: 4px;
    border-radius: 4px;
}
QPushButton[variant="icon"]:hover { background: rgba(255,255,255,13); }
QPushButton[variant="icon"]:pressed { background: rgba(255,255,255,20); }

/* --- QPushButton (Segmented / Toggle group) --- */
QPushButton[variant="segment"] {
    background: #0E0F14;
    color: rgba(255,255,255,102);
    border: 1px solid rgba(255,255,255,51);
    border-radius: 6px;
    padding: 8px 16px;
    font-family: "Inter";
    font-size: 12px;
}
QPushButton[variant="segment"]:hover {
    background: rgba(255,255,255,10);
    color: rgba(255,255,255,153);
}
QPushButton[variant="segment"]:checked {

```

```

        background: #3B82F6;
        color: #FFFFFF;
        border-color: #3B82F6;
        font-weight: 600;
    }

/* --- QProgressBar (Health Bar) --- */
QProgressBar {
    background: rgba(255,255,255,20);
    border: none;
    border-radius: 2px;
    max-height: 4px;
    min-height: 4px;
    text-align: left;
}
QProgressBar::chunk {
    border-radius: 2px;
    background: #22C55E;
}
QProgressBar[health="warn"]::chunk { background: #F59E0B; }
QProgressBar[health="bad"]::chunk { background: #EF4444; }

/* --- QSlider (Volume / Opacity) --- */
QSlider::groove:horizontal {
    height: 4px;
    background: rgba(255,255,255,20);
    border-radius: 2px;
    border: none;
}
QSlider::sub-page:horizontal {
    background: #3B82F6;
    border-radius: 2px;
}
QSlider::handle:horizontal {
    width: 12px;
    height: 12px;
    background: #FFFFFF;
    border-radius: 6px;
    margin: -4px 0;
    border: 2px solid #3B82F6;
}
QSlider::handle:horizontal:hover { background: #3B82F6; }
QSlider::groove:horizontal[thin="true"] { height: 3px; }
QSlider::handle:horizontal[thin="true"] { margin: -4px 0; }

/* --- QLabel (Section Header) --- */
QLabel[role="section-header"] {
    color: rgba(255,255,255,102);
    font-family: "Inter";
    font-size: 10px;
    font-weight: 600;
    letter-spacing: 1px;
    text-transform: uppercase;
}

/* --- QLabel (Monospace data) --- */
QLabel[role="mono"] {
    font-family: "JetBrains Mono";
    font-size: 10px;
    color: rgba(255,255,255,89);
}

/* --- QLabel (Page title) --- */
QLabel[role="page-title"] {

```

```

        font-family: "Space Grotesk";
        font-size: 12px;
        font-weight: 700;
        color: #FFFFFF;
        letter-spacing: 1px;
        text-transform: uppercase;
    }

/* --- QTableWidget (Sessions Table) --- */
QTableWidget {
    background: #111318;
    border: none;
    gridline-color: rgba(255,255,255,18);
    color: rgba(255,255,255,153);
    font-family: "Inter";
    font-size: 12px;
}
QTableWidget::item:selected {
    background: rgba(59,130,246,26);
    color: #FFFFFF;
}
QTableWidget::item:hover { background: rgba(255,255,255,20); }
QHeaderView::section {
    background: #0E0F14;
    color: rgba(255,255,255,77);
    font-family: "Inter";
    font-size: 10px;
    font-weight: 600;
    letter-spacing: 1px;
    border: none;
    border-bottom: 1px solid rgba(255,255,255,18);
    padding: 8px 12px;
}

/* --- QScrollArea --- */
QScrollArea {
    background: transparent;
    border: none;
}
QScrollArea > QWidget > QWidget { background: transparent; }

/* --- QFrame (Panel border) --- */
QFrame[role="panel"] {
    background: #111318;
    border: 1px solid rgba(255,255,255,20);
    border-radius: 8px;
}

/* --- QFrame (Divider) --- */
QFrame[role="divider-v"] {
    background: rgba(255,255,255,26);
    max-width: 1px;
    min-width: 1px;
    border: none;
}
QFrame[role="divider-h"] {
    background: rgba(255,255,255,18);
    max-height: 1px;
    min-height: 1px;
    border: none;
}

/* --- QToolTip --- */
QToolTip {

```



```
background: #16181F;
color: rgba(255,255,255,204);
border: 1px solid rgba(255,255,255,31);
border-radius: 4px;
font-family: "Inter";
font-size: 11px;
padding: 4px 8px;
}
```

17. Responsive Rules

The prototype was designed at 1440×900. The following rules govern behavior at smaller sizes.

17.1 Minimum Window Size

```
window→setMinimumSize(1024, 706);
```

Below 1024×706, show a “Window too small” overlay or disable resize below this threshold.

17.2 Dashboard at 1024×706

At 1024px wide, the available main content width is $1024 - 220 \text{ (sidebar)} = 804\text{px}$.

Panel	1440px behavior	1024px behavior
Stream Status panel	240px fixed	220px fixed (reduce by 20px)
Center column	flex-1 (980px)	flex-1 (584px)
Scene cards	all 6 visible	horizontal scroll activates at ~4 visible
Audio mixer	4-column grid	4-column grid maintained (cards narrow)
Program output	16:9 aspect	16:9 aspect maintained, height reduces

17.3 Scene Editor at 1024×706

Panel	1440px behavior	1024px behavior
Right panel (Sources + Properties)	320px fixed	280px fixed
Canvas workspace	flex-1 (900px)	flex-1 (524px)
Properties panel	full scroll	full scroll (same)
Transitions bar	all 5 buttons + duration	all 5 buttons + duration (tighter spacing)

17.4 General Rules

- Sidebar never collapses** — always 220px at all supported sizes.
- Top bar never wraps** — at narrow widths, right-side elements may be clipped. Priority: LIVE badge > timecode > resolution/fps/codecs > SIG OK > profile.
- Panels never overlap** — use `QSplitter` for resizable panels if desired, but default to fixed widths.
- Minimum panel widths:** Center column min-width 400px; right panels min-width 200px.
- Font sizes never scale** — all sizes are fixed pixel values, not em/rem.
- Scroll areas activate** when content overflows — never clip content silently.

Document version: 1.0 — Generated from Nexus Broadcast UI Prototype (React/Tailwind) for Qt 6 Widgets C++ implementation.
Prototype source: `nexus-broadcast-ui-source.zip` Icon sources: `nexus_icons.zip` (54 Lucide SVG files)